SEMplex browser application user guide

Version 1.3 · https://s-matysik.github.io/semplex/

1. What is SEMplex

SEMplex guides you through the entire analysis of survey data: from importing a file, cleaning the data, assessing scale reliability, and conducting factor analysis (EFA, CFA), to structural equation modeling (SEM), mediation, and an APA-style report. Its distinctive feature is **Approach comparison**: it estimates the same relationship using seven methods on the same data and shows whether the conclusion depends on the method. The **Decision log** records every change to the data and model and lets you undo it.

- **Your data stay on your computer.** The browser reads the file, and a Python engine running in the same tab (WebAssembly) performs the calculations. Nothing is sent to a server.
- **No installation or account required.** Simply open the website.
- **The application suggests; the researcher decides.** Suggestions, such as reverse-coding an item, excluding respondents, or modifying a model, require your approval. Each decision is recorded in the log.

2. First launch

1. Open https://s-matysik.github.io/semplex/ in an up-to-date desktop browser (Chrome, Edge, or Firefox). Phones and tablets are not recommended because the results screens are wide.
2. On first launch, the application downloads the Python environment (approximately 25 MB). You will see **Starting the computation engine** with a progress bar. This usually takes 20–40 seconds. On subsequent launches, the browser uses its cache, so startup is faster.
3. Once the engine is ready, the **Data and cleaning** module opens.

Screen layout

Element	Purpose
Sidebar	Lists the modules in analysis order: Data and cleaning → Descriptives and tests → Reliability and validity → EFA → CFA → SEM – model editor → Mediation and moderation → Approach comparison → Report → Decision log. Modules marked as planned (Group comparison, Power and sample size, PLS-SEM, Multilevel models) do not yet perform calculations.
Analysis progress bar (top)	Shows the steps Data → Reliability → EFA → CFA → SEM → Report and their status (completed, current, pending, out of date).
Mode: Simple / Advanced	Simple mode hides some technical details, such as code export. For research, select **Advanced**.
Gear icon	Settings: random seed, number of bootstrap resamples, and report language (Section 13).
Moon icon	Light / dark theme.
Question mark icon	Help and keyboard shortcuts.
Details panel (right)	Click an item, path, or table row to view details and suggested actions. In a narrow window, use the Details button to open the panel.
Task tray	Lengthy computations, such as bootstrapping, run in the background with a progress bar and can be canceled.

Keyboard shortcuts

Shortcut	Action
Ctrl K	Command palette: quickly navigate to a module or command, such as “Settings” or “Open project”.
Ctrl S	Save the project (.sxp) without columns omitted during import.
Ctrl Enter	In the CFA/SEM syntax editor: run the model.

3. Preparing the data file

- Formats: **CSV, TXT, TSV, XLSX**. The encoding (UTF-8, Windows-1250, ISO 8859-2), delimiter, and decimal separator are detected automatically.
- One row = one respondent; one column = one item (question). The first row contains the column names.
- Name items using a scale prefix and a number, such as SAT1, SAT2, SAT3, LOY1… This allows SEMplex to suggest scales automatically.
- Responses may be numeric (1–5, 1–7, 0–10) or verbal Likert-scale labels (“somewhat agree”). Verbal labels are converted to numbers.
- Missing data: empty cells are always treated as missing. Specify any additional codes, such as −99, “don't know”, or “no response”, during import.
- Columns containing personal data (email, phone number, PESEL national identification number, first name) are detected and suggested for omission.

Tip. Want to practice first? On the import screen, click **Open example: Customer loyalty 2026 (N = 524)**. Section 15 provides a 10-minute walkthrough of this example.

4. Step 1 – Data and cleaning

4.1. Importing a file

1. In the **Data and cleaning** module, drag your file onto **Drop a data file here or select one from your computer**, or click **Select file**.
2. Check the **Preview of the first rows**. If Polish characters display incorrectly and you see **Some characters do not match the encoding**, change **Encoding** to Windows-1250. You can also set the **Delimiter** and **Decimal separator** manually. The preview updates immediately.
3. In **Missing-value codes**, add codes that indicate a missing response (type a code and press Enter), such as 99 or 0 if 0 means “not applicable”.
4. In the variable table, check the **Type** of each column: ordinal (Likert scale), continuous, binary, nominal, or omit. Columns that identify individuals are flagged and omitted by default.
5. Click **Import N variables**.

4.2. Overview tab

- A data table shows the number of respondents, missing values, and complete cases.

- Click a column heading to view the item's distribution and statistics. If an item correlates negatively with the rest of its scale, the application suggests **Reverse-code item** and shows how the scale's α would change. Reverse-coding is recorded in the log.

4.3. Careless respondents tab

1. If the data include completion time, select the relevant variable in **Completion time (seconds)**.
2. In **Who to flag**, select a rule: **both signals (recommended)**—identical responses across many consecutive items (longstring) and an unusually short completion time; **either signal**; or **none – longstring only**. Multivariate outliers are also flagged using Mahalanobis distance.
3. Review **Suggested exclusions** and click **Exclude N from analysis**. Exclusions are reversible: the rows remain in the data with a flag, and you can undo the decision in the log.

4.4. Missing data tab

Shows the items with the most missing values, **Most common combinations of missing items**, and Little's MCAR test. Note: models (CFA, SEM, approach comparison) use complete cases, so missing values scattered across many items reduce the analysis sample.

4.5. Scales tab

1. Check the scales suggested from the item names.
2. Edit them in **Scale definitions**. Enter one scale per line in the format ABBREVIATION (name): comma-separated items, for example, SAT (Satisfaction): Q1, Q2, Q3.
3. Click **Save scales**. Changes to scales are recorded in the log and mark dependent results for recalculation.

5. Step 2 – Descriptives and tests

- Item table: mean, standard deviation, **Skewness**, **Kurtosis**, missing values, and a small distribution plot.
- **Correlation matrix**: Pearson or Spearman correlations.
- **Compare groups by**: select a grouping variable, such as gender or branch. The application calculates Welch's t-test with Hedges' g, or ANOVA, for scale means.
- Mardia's test of multivariate normality guides the choice of **Estimator** for CFA and SEM. For nonnormal distributions, select **MLM (Satorra–Bentler)** for robust standard errors and a scaled χ^2. This choice applies to both CFA and SEM.

6. Step 3 – Reliability and validity

- Scale reliability: Cronbach's α (with a Feldt confidence interval), McDonald's ω, and item-deleted statistics. **Bootstrap ω and α** calculates bootstrap confidence intervals.
- **Sample**: **Full sample** or **CFA subsample** (when the sample is split in EFA).
- Convergent validity from the measurement model for all scales: CR, AVE, and the range of loadings λ.
- Discriminant validity: a matrix with factor correlations below the diagonal, HTMT above it, and √AVE on the diagonal (the Fornell–Larcker criterion), plus MSV.
- Each value has a verbal assessment and a threshold with a cited source, for example, $\alpha \geq 0.70$, AVE ≥ 0.50, and HTMT < 0.85.

7. Step 4 – EFA (exploratory factor analysis)

1. **Sample split:** leave **Split the sample into EFA and CFA subsamples** enabled (recommended: confirm the structure identified by EFA using different respondents in CFA). Set the **Split seed** and, optionally, **Strata**, such as gender. Without a split, CFA fit is inflated. Disable the split if subsequent models (SEM, approach comparison) should use the full sample. This change is recorded in the log.
2. Select **Correlations** (Pearson or polychoric, which are better suited to 4–5-point scales), **Extraction** (minres (ULS), maximum likelihood, or principal axis factoring (PAF)), and **Rotation** (oblique: oblimin, geomin, or promax, which allow factors to correlate; orthogonal: varimax or quartimax).
3. Check **KMO** and **Bartlett's test** to assess whether the data are suitable for factor analysis.
4. Choose the number of factors using the **Scree plot with parallel analysis** or the **Criteria consensus** list. The list includes nine criteria, such as parallel analysis, MAP, BIC, and the Kaiser criterion, and provides a weighted recommendation. The decision is yours: you may retain the number of factors supported by theory. **Return to consensus** restores the recommendation.
5. Review **Factor loadings** and **Factor correlations (Φ)**. Items with weak loadings or cross-loadings are flagged. You can select **Omit from CFA** for these items and click **Recalculate for selected items**.
6. Optionally, select **Check stability** to bootstrap the loadings.
7. Click **Build CFA model**. The application generates the measurement-model syntax and opens CFA.

8. Step 5 – CFA (confirmatory factor analysis)

1. In **lavaan syntax**, review or enter the model (see the syntax reference in Section 16) and press Ctrl Enter.
2. Set **Estimator** (ML or MLM) and **Missing data**: **Complete cases only (listwise)** or **EM (covariance)**. EM mode has not been validated; use it with caution. The change is recorded in the log.
3. Assess model fit: χ^2 (scaled for MLM), CFI, TLI, RMSEA with a 90% CI, and SRMR. Each index has a verbal assessment and a threshold.
4. Results tabs: **Loadings**, **Correlations**, **Residuals and R^2**, **Modification indices**, and **Model comparison**.
5. **Modification indices:** the **Add** button next to each suggestion (MI, EPC, sEPC) adds it to the model. Make only substantively justified changes, such as a residual covariance between items with similar wording. Each change is recorded in the log.
6. To compare variants, click **Save for comparison** for each model, then open **Model comparison → Compare saved models**. The scaled $\Delta\chi^2$ test is calculated only for nested models using the same variables and rows. Otherwise, the application compares ΔCFI, AIC, and BIC.
7. **Bootstrap CI** calculates bootstrap confidence intervals for standardized parameters.
8. Warnings, such as **Inadmissible solution (Heywood case)**—a negative variance or a loading > 1—indicate what to check in the model.
9. In **Advanced** mode, ready-to-use lavaan and Python code appears below the results, along with **Analysis data (CSV)** for use with the code. The file contains exactly the rows and variables used to estimate the model, allowing you to reproduce the result in R.

9. Step 6 – SEM (model editor)

1. Build the model on the canvas or in the **Syntax** tab. Changes on the canvas and in the syntax stay synchronized.

2. **Model tools: Select**, **Latent variable** (a new variable with items), **Path** (regression), **Covariance**, and **Measurement** (loading). Delete a selected element using **Delete relation / Delete variable**. **Undo** and **Redo** work as they do in a text editor.
3. In the syntax, specify paths as Y ~ X1 + X2. Define indirect effects using labels, for example, M ~ a*X, Y ~ b*M, ind := a*b. They appear in the **Defined effects** table.
4. Structural results: standardized paths with 95% CIs, R^2, and model fit. **Bootstrap CI** provides intervals for paths and effects.
5. The application warns about collinearity: latent correlations ≥ 0.90 and standardized coefficients outside [−1, 1]. These signals indicate that the separate effects of correlated predictors should not be interpreted.
6. Next to a path, **Approach comparison** (**Compare this effect using other approaches**) transfers it to the comparison module (Section 11).

10. Step 7 – Mediation and moderation

1. Select a **Model template** (equivalent to PROCESS templates): **Template 4 · mediation** (X → M → Y), **Template 1 · moderation** (W changes the strength of X → Y), **Template 7** (W moderates path a), or **Template 14** (W moderates path b).
2. Select the scales: X – predictor, M – mediator, W – moderator, Y – dependent variable. In Template 4, you can enable **Latent variables (SEM)** instead of scale means.
3. Read the results: **Indirect effect a·b** with a bootstrap confidence interval (**Bootstrap indirect effect**), **Direct effect c'**, **Total effect**, and **Proportion mediated**; for moderation, **Simple-slope probes** and the **Johnson–Neyman region**; for moderated mediation, the **Index of moderated mediation**.

11. Step 8 – Approach comparison (key feature)

This module answers the question: does the result depend on the method? It estimates the same structural relationship using seven approaches, with the same cases, scales, and bootstrap resamples. Differences therefore arise only from how the constructs are represented.

Approach	What it does
Regression on item means	Represents constructs as item means; ignores measurement error, usually underestimating effects.
Factor scores (regression method)	Regresses on factor scores; may overestimate effects.
Croon's correction	Uses factor scores with a correction for measurement error.
Single indicator with α correction	Uses the scale mean with error variance fixed based on reliability.
SAM (local)	First estimates each scale's measurement model separately, then the structural model; robust to misspecification in a single scale.
ESEM approximation	Allows cross-loadings (target rotation).
Joint SEM	Estimates measurement and structural components jointly; serves as the reference.

1. Select the **Dependent variable** and the first predictor under **Predictors**. Use **Add predictor** to add further predictors; each has its own selection field.

2. Click **Compare 7 approaches**. With the default 200 bootstrap resamples, this usually takes 1–3 minutes. Progress appears in the task tray, and preliminary results are displayed during computation.
3. With multiple predictors, **Show effect** selects which effect to display.
4. Read the **Forest plot of the effect across seven approaches**: points represent standardized β coefficients, lines show 95% confidence intervals, and the shaded band shows the joint SEM interval.
5. In the table, pay attention to **Δ vs SEM** (the difference from joint SEM) and **Admissible resamples** (the number of bootstrap resamples that produced an admissible solution). An interval is reported only when at least 50 resamples are admissible and they account for at least 80% of all resamples.
6. The **Successive differences between approaches** panel shows how the estimate changes step by step. This is descriptive, not a decomposition of causes.
7. Read the application's recommendation. It is a heuristic rule, such as SAM for N < 300, rather than an indication of the “true” result. Also check the warning about predictor collinearity ($|r| \geq 0.85$).
8. Copy the ready-to-use paragraph for the **Methods** section (**Paragraph language**: Polish or English).

Interpreting the result

- All approaches yield the same sign, and their intervals exclude zero: the conclusion is robust to the choice of method.
- The sign changes across approaches, or the intervals are very wide: do not interpret this predictor's separate effect; report the result's sensitivity. Strongly correlated predictors are usually the cause.
- Estimates based on item means are clearly below those from latent-variable approaches: measurement error attenuates the effect.
- Few admissible resamples in joint SEM: the model is unstable in this sample.

12. Step 9 – Report and Decision log

12.1. Report

1. Select **Report language** (Polish or English, APA 7).
2. Select the sections: EFA, Reliability, CFA, SEM, Mediation, Approach comparison, and Decision log.
3. If you see **Some results are out of date**, rerun the indicated modules before exporting.
4. Export as **HTML**, **Word**, or **Print / PDF**. The report contains a results narrative and APA-style tables.

12.2. Decision log

- A list of changes: what changed, why, and with what effect (before → after), including the source (manual, automatic, or suggested by the application). Options include filters, **Search the log**, and **Show undone decisions**.
- **Researcher's note**: add a rationale for a decision that the application does not record automatically, such as retaining the theoretical number of factors.
- **Undo this decision…**: the application shows the **Dependent modules** whose results will need to be recalculated. The entry remains in the history with strikethrough formatting.
- **Open module** takes you to where the decision was made. Export the log as CSV or JSON, for example, as supplementary material for an article.

When a module is marked **Results are out of date** after changes to the data or scales, or after a decision is undone, click **Recalculate now**.

13. Settings

Setting	Default	Meaning
Random seed	1	Produces the same bootstrap results and sample split each time the analysis is run.
SEM and mediation bootstrap	500	Number of resamples for confidence intervals for paths and indirect effects.
Approach comparison bootstrap	200	Each resample runs all 7 approaches. Using 500 gives more stable intervals but takes approximately 2.5 times longer.
EFA loading stability	100	Number of resamples for assessing EFA stability.
Report language	Polish	Polish or English (APA).

14. Saving and opening a project

- **Ctrl S** (or **Save project**) saves an .sxp file containing the data, settings, models, results, and log. By default, the file excludes columns omitted during import, such as personal data, but includes the remaining responses. Store it with the same care as the data themselves.
- To save the **full source file**, including all columns and personal data, use the Ctrl K command palette and select **Save project with full source file (including personal data)**.
- **Open project** (in the Data module, or **Open project (.sxp)** in the Ctrl K command palette) restores your work, including the import settings.
- The application does not store data on a server. Closing the tab without saving means losing any unsaved work.

15. Quick start in 10 minutes (example data)

1. Data and cleaning → **Open example: Customer loyalty 2026 (N = 524)**. The email column is recognized as personal data and omitted.
2. Overview tab → click the heading for Q6 ("I often think about switching providers") → **Reverse-code item**.
3. Careless respondents tab → select **czas_s** as the completion-time variable → **Exclude N from analysis**.
4. Scales tab: SAT (Q1–Q3), LOY (Q4–Q6), TRU (Q7–Q9), VAL (Q10–Q12) → **Save scales**.
5. Reliability and validity → check α, ω, and HTMT.
6. EFA (sample split enabled) → select the number of factors → **Build CFA model**.
7. CFA → assess fit and review the modification indices.
8. SEM → in the syntax, add, for example, **LOY ~ SAT + TRU + VAL**, then press Ctrl Enter.
9. Approach comparison → dependent variable LOY, predictor SAT (and, for example, TRU) → **Compare 7 approaches**.
10. Report → **Word**. Decision log → review the recorded steps.

16. Model syntax reference (lavaan)

Notation	Meaning	Example
=~	latent variable measured by items	SAT =~ sat1 + sat2 + sat3
~	regression (path)	LOY ~ SAT + IMAG
~~	covariance (including residual covariance)	expe1 ~~ qual1
:=	defined effect	ind := a*b
a*X	parameter label	M ~ a*X
1*X	fixed parameter	F =~ 1*x1 + x2 + x3
#	comment	# measurement model

The first loading of each latent variable is fixed to 1 by default. Covariances between exogenous variables are added automatically.

17. Troubleshooting

Problem	What to do
The computation engine failed to start	Refresh the page. An internet connection is required on first launch to download the environment. Check that your browser is up to date; caching may be disabled in private browsing mode.
Garbled Polish characters	During import, change Encoding to Windows-1250.
All values appear in one column	Change the Delimiter (semicolon, comma, or tab).
Decimal numbers are treated as text	Change the Decimal separator (comma / period).
Small sample in CFA or SEM	Models use complete cases. Check the Missing data tab, the missing-value codes, and whether the sample has been split (EFA → sample split).
Inadmissible solution (Heywood case)	Negative variance or a loading > 1: check for items with weak loadings, an insufficient sample size, or scales with only two items.
Coefficients |β| > 1 and a collinearity warning	The predictors are too strongly correlated. Report their combined effect, check discriminant validity (HTMT), and use Approach comparison.
Results are out of date	Click Recalculate now in the relevant module.
Bootstrapping takes a long time	Reduce the number of resamples in Settings or cancel the task in the task tray.
This screen encountered an error	Switch to another module and return; save the project (Ctrl S) and refresh the page.

18. Limitations of version 1.3

- ML and MLM estimators; no MLR or WLSMV for ordinal data (planned).
- Missing data: complete cases; EM mode in CFA has not been validated; no FIML.
- No mean structure or multigroup comparisons (invariance testing is planned).
- ESEM in approach comparison is an approximation; moderated mediation templates use scale means.
- The engine's agreement with lavaan, semTools, and psych was tested on 1,988 numerical values. Validation applies to the models tested.

Source code, documentation, and bug reports: https://github.com/s-matysik/semplex